

The University of Zambia
Department of Mathematics and Statistics
Mat 1110 - Foundation Mathematics and Statistics for
Social Sciences

Tutorial Sheet 10- Derivatives and its applications

September, 2023.

1. Find the second derivative in each of the following:

(a) $y = \sqrt{x^2 + 1}$ (b) $y = \pi x^3 - 7x$ (c) $y = \frac{1}{x^2+1}$ (d) $y = \frac{x+1}{x-1}$

2. Show that if

(a) $y = 2 \sec^3 x$, then $\frac{dy}{dx} = 6 \sec^3 x \tan x$ (d) $y = \cos^{-1} x$, then $\frac{dy}{dx} = \frac{-1}{\sqrt{1-x^2}}$.
(b) $y = \sqrt[4]{x^2 + 1}$, then $\frac{d^2 y}{dx^2} = \frac{2-x^2}{4(x^2+1)^{\frac{7}{4}}}$. (e) $y = \tan^{-1} x$, then $\frac{dy}{dx} = \frac{1}{1+x^2}$.
(c) $y = \sin^{-1} x$, then $\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$.

3. Find the equation of the tangent and of the normal to each of the following functions at a given value of x .

(a) $f(x) := \sqrt{x^2 + 16}$ at $x = 3$. (c) $f(x) := (\ln x)^2$ at $x = 1$.
(b) $f(x) := \frac{x+3}{x-4}$ at $x = -3$.

4. Find the point at which the gradient of the curve

$$f(x) := 5 + 3x - x^3 \text{ is } 3.$$

5. Find the point at which the gradient of the curve

$$f(x) := \sin x \text{ is } -1, \text{ for } 0 \leq x \leq 2\pi$$

6. Find the point at which the gradient of the curve

$$f(x) := \cos x \text{ is } \frac{1}{2}, \text{ for } 0 \leq x \leq 2\pi$$

7. Find the critical numbers and determine whether they yield a relative maxima or relative minima or inflection point **using the second derivative test**, and hence sketch the graph.

(a) $f(x) = 2x^2 - 4x + 5$ (d) $f(x) = x^3 - 5x^2 - 22x + 56$
(b) $f(x) = -2x^2 - 3x + 2$ (e) $f(x) = x^3 - 5x^2 + 3x + 9$
(c) $f(x) = 4x^3 + 9x^2 - x - 6$ (f) $f(x) = x^4 + 4x^3 - 12x^2$

8. Find the maxima and minima of a function $y = 2x^3 - 3x^2 + 6$ using derivatives.
9. Suppose the demand equation for a monopolist is $p = 100 - 0.01x$ and the cost function is $C(x) = 50x + 10000$. Given that the total revenue function is given by $R(x) = xp$, where p is the demand equation as given above. find the value of x that maximizes the profit and determine the corresponding price and total profit for this level of production. [Hint: Profit function is given by $P(x) = R(x) - C(x)$].
10. Rework question 9 under the condition that the government has imposed an excise tax of $ZMK10$ per x unit.
11. The revenue function for a particular production is $R(x) = x(4 - 0.0001x)$. Find the largest possible revenue.
12. Let $P(x) = -0.01x^2 + 550x - 10,000$ be the profit function for a monopolist. Find the value of x that maximizes the profit and determine the corresponding profit.
13. A printer needs to make a poster that will have a total area of $200m^2$ and will have $1m$ margins on the sides, a $2m$ margin on the top and a $1.5m$ margin on the bottom. What dimensions of the poster will give the largest printed area?
14. A window is being built and the bottom is a rectangle and the top is a semicircle. If there is $12m$ of framing materials what must the dimensions of the window be to let in the most light?